Notes

Pierson Lipschultz

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Contents

Chapter 1

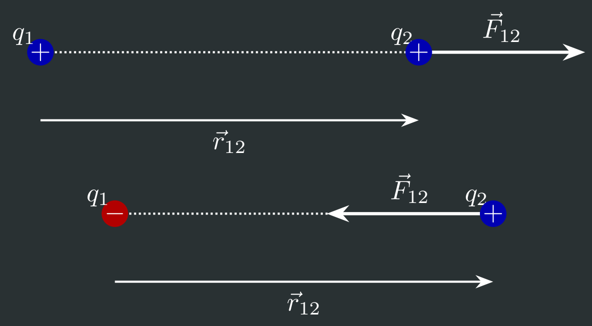
class notes

1.1 Coulomb's Law

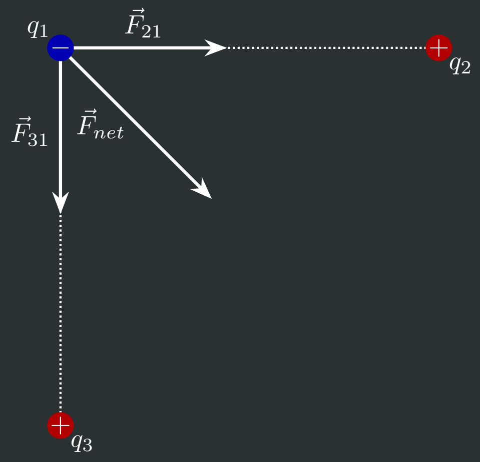
Coulomb's Law

$$\vec{F_{12}} = k\frac{q_1 q_2}{r_{12}^2}\hat{r_{12}} \tag{1.1}$$

Gives a force along the line of two charges.
If q_1 and q_2 have the same sign (i.e. charge) then the force must be repulsive. But if they have the *same* sign, then it must be attractive.
Because of Newton's second law, $F_{12} = -F_{21}$



$$\boxed{F_{1,net} = F_{21} + F_{31}}$$



Superpostion Principle

$$F_{net} = \sum_i \vec{F_i} \tag{1.2}$$

Electron Charge

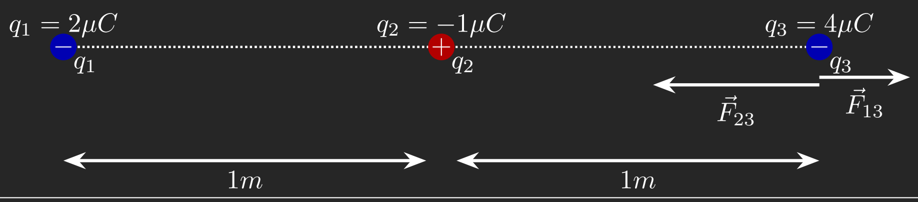
$$e = -1.6 \times 10^{-19} C$$

k

$$k = 9 \times 10^9 \frac{NM^2}{C^2}$$

1.1.1 Examples

Example 1.1.1 $\boxed{F_{3,net} = ???}$



Using Coulomb's Law1.1...

$$\vec{F_{3,net}} = \vec{F_{31}} + \vec{F_{32}}$$

$$\vec{F_{31}} = k\frac{q_3 q_1}{r_{31}^2}\hat{r_{32}}$$

$$\vec{F_{32}} = k\frac{q_3 q_2}{r_{32}^2}\hat{r_{32}}$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2}, \qquad q_1 = 2\mu C, \qquad q_2 = -1\mu C, \qquad q_3 = 4\mu C$$

$$\vec{F_{31}} = 1.800 \times 10^{-2}\,\mathrm{N}, \quad \vec{F_{32}} = -3.600 \times 10^{-2}\,\mathrm{N}$$

$$\boxed{\vec{F}_{3,net} = -1.800 \times 10^{-2}\,\mathrm{N}}$$

1.2 Electric flux and Field Lines·

1.2.1 Field Lines

- Measure of electric field lines over an area

Electric Field

$$E = k\frac{q}{r^2} \tag{1.3}$$

Measure of the energy in an electric field at a point

Density of Field Lines

$$D = \frac{N}{4\pi r^2} \tag{1.4}$$

used when plotting

Number of Graphed Field Lines

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \tag{1.5}$$

1.2.2 Flux

Number of field lines passing through a surface

$$\Phi \propto N$$

For a point charge aligned at the center at a sphere it can just be defined as

$$\Phi \equiv EA$$

To calculate total over surface ...

$$\Phi \equiv \int_s \vec{E} d\vec{A}$$

1.3 5: potential energy

Guass's Law

$$\int EdS = \frac{Q_{enclosed}}{\epsilon_0} \tag{1.6}$$

Change in potential in const E field

$$\Delta U = q\vec{E}\ell \tag{1.7}$$

Change in potential in non static field

$$\Delta U = Q \int_a^b \vec{E}dr \tag{1.8}$$

1.7 Conductors and Capacitance

1.7.1 Conductors

- Every point in a conductor is at equipotential, this is because the electric field at any point is zero.
- Contuining with Gausses law, no matter the location of a charge contained *inside* a shell (but not overlapping the shell itself), the outside of the shell will have the same charge distribution, however the inside may not.
- When looking at the inverse, (a particle placed *outside* of a conductor), we find that the charges move to the surface to cancel out the exerted field, meaning that the inside of the object will have a net charge of 0.
- Importantly, this still applies when a cavity is made *within the conductor*
- This allows for the shielding of an object if it is placed within a conductor [1]

[1] Think Faraday cage (maybe)

Example 1.7.1 Equipotential difference in two Spheres

$$V_A = -\int_{\infty}^{R_A} \vec{E}_A \cdot \vec{dl}, \qquad V_B = -\int_{\infty}^{R_B} \vec{E}_B \cdot \vec{dl}$$

Given a point charge ...

$$\vec{E} = k\frac{Q}{r^2} \rightarrow -\int_{\infty}^{r} k\frac{Q}{r^2} = k\frac{Q}{r} \rightarrow$$

For a point charge...

$$V = k\frac{Q}{r}$$

$$V_A = k\frac{Q}{R_A}, \qquad V_B = k\frac{Q}{4R_B}$$

$$4 \times V_A = V_B$$

Once connected, become a single conductor which wishes to reach equipotential. To find final charges ...

$$V_A = V_B \rightrightarrows k\frac{Q_A}{R_A} = k\frac{Q_B}{R_B}$$

$$\cancel{k}\frac{Q_A}{R_A} = \cancel{k}\frac{Q_B}{4R_A}$$

$$4Q_A = Q_B$$

1.7.2 Capacitance

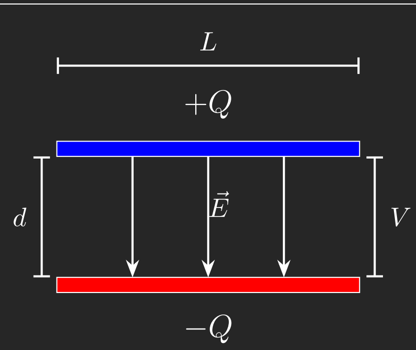
Capacitance

$$C \equiv \frac{Q}{\Delta V} \tag{1.9}$$

Units of Farads, or Coulombs per volt

- Measure of field between to oppositely charged objects
- Ex. A plate of Q and -Q placed a distance d away from one another
- Stores the energy *in* the magnetic field induced between said objects

Example 1.7.2 Capacitance between two plates



$$\boxed{d << L}$$

To calculate E field between plates . . .

$$E_{bot} = \frac{1}{2}\frac{\sigma}{\epsilon_0}, \quad E_{top} = \frac{1}{2}\frac{\sigma}{\epsilon_0}$$

$$\sigma = \frac{Q}{A}$$

$$E_{bot} = \frac{1}{2}\frac{Q}{\epsilon_0 A}, \quad E_{top} = \frac{1}{2}\frac{Q}{\epsilon_0 A}$$

Since field lines are the same direction. . .

$$E_{tot} = \frac{Q}{\epsilon_0 A}$$

To find capacitance. . .

$$C \equiv \frac{Q}{\Delta V}$$

$$|\Delta V| = \int_{bot}^{top} \vec{E} \cdot dl \rightrightarrows \Delta V = \frac{Q}{\epsilon_0 A}\int_{bot}^{top} dl$$

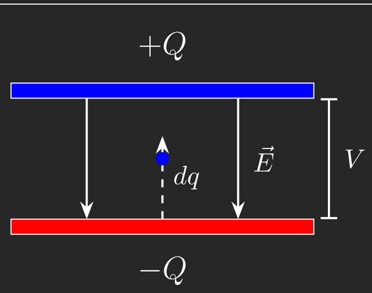
$$\Delta V = \frac{Q}{\epsilon_0 A} d$$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance for Parrel Plates

$$C = \frac{\epsilon_0 A}{d} \tag{1.10}$$

Example 1.7.3 Parallel Plate Capacitor



To calculate the work needed to move a particle from the bottom plate to the top plate

$$dW = dqEd$$

Since we know that the energy from the electric field is equal to $\frac{V}{d}$...

$$dW = dq\frac{V}{\cancel{d}}\cancel{d}$$

$$dU = V dq$$

This holds true for *any system.* Note that dU is *not* constant because for each particle moved, it becomes requires more work for the next.
To calculate potential of a capacitor...

$$U = \int_0^Q V dq \Rrightarrow U = \int_0^Q \frac{q}{c} dq$$

Thus ...

$$U = \frac{1}{2}\frac{Q^2}{C}$$

Potential of a Capacitor

$$U = \frac{1}{2}\frac{Q^2}{C} \quad U = \frac{1}{2}CV^2 \quad U = \frac{1}{2}QV \tag{1.11}$$

Energy density of a capacitor

$$u = \frac{1}{2}\epsilon_0 E^2 \tag{1.12}$$

1.8 Capacitors

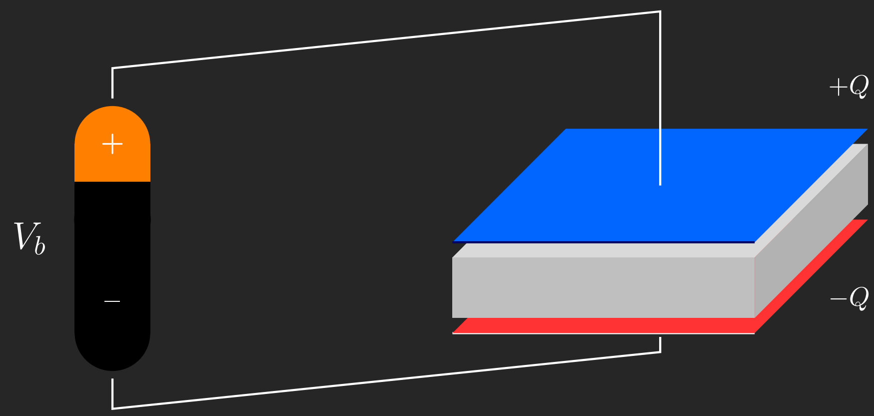
What happens if we pull apart a parallel plate capacitor?

- Q remains constant
- E increases
- ΔV increases
- C decreases
- U increases

1.8.1 Dielectrics

- Material where positive and negative charges in individual molecules can change, but no through the entire material
- When placed between a capacitor all of the charges align themselves to the field
- This creates a material full of little dipoles
- This induced field is the *opposite* of the one from the plates
- This has the effect of *reducing* the electric field within the capacitor, which then reduces the potential diff (V), which then *increases* the capacitance
- $$C = \frac{Q}{V}$$
- The amount a capacitance increases is called the Dielectric Constant and is calculated via $C_{old} = \kappa C_{new}$

Example 1.8.1 Adding a Dielectric to a Capacitor



- Capacitance increases
- $V_c = V_b$
- Because $C = \frac{Q}{V} \rightrightarrows Q \uparrow$
- Because $U = \frac{1}{2}QV_c \rightrightarrows U \uparrow$

1.8.2 Capacitors in Parallel

- When wired in parallel they can be treated as one capacitor with a larger area
- this equivalent capacitor has...

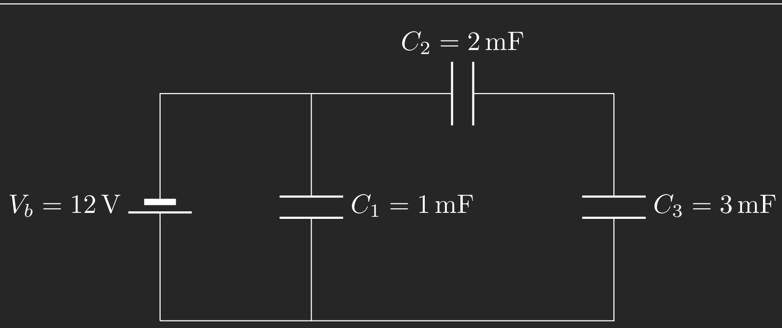
$$V_{equiv} = V_1 = V_2 \qquad C_{equiv} = C_1 + C_2, \qquad Q_{equiv} = Q_1 + Q_2$$

1.8.3 Capacitor in series

- When wired in series they act like a capacitor with a large distance
- This equivalent capacitor has ...

$$Q_{equiv} = Q_1 = Q_2 \qquad \frac{1}{C_{equiv}} = \frac{1}{C_1} + \frac{1}{C_2}, \qquad V_{equiv} = V_1 + V_2$$

Example 1.8.2 Combination of Capacitors



Simply into single capacitors

$$C_2, C_3 \to \frac{1}{C_{23}} = \frac{1}{C_2} + \frac{1}{C_3} = 1.2\text{mF}$$

$$C_1, C_{23} \to C_1 + C_{23} = C_{123} = 2.2\text{mF}$$

1.9 Kirchhoff's Rules

Kirchhoff Voltage Rule

$$\sum \Delta V_n = 0 \tag{1.13}$$

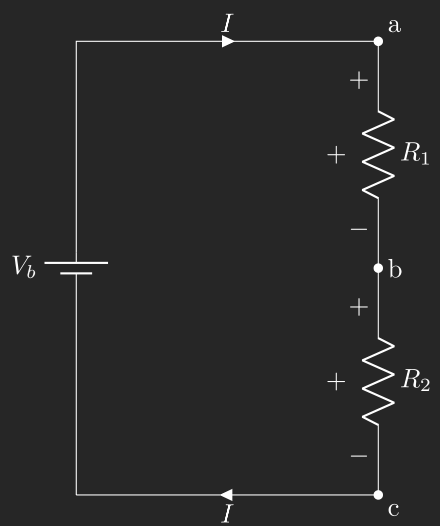
In a ***closed*** circuit, if you return to the same location, no matter what path you take, the change in voltage will be 0. Very similar to conservation of energy in a magnetic field.

Kirchhoff Current Rule

$$\boxed{\sum I_{in} = \sum I_{out}}\text{, at any node} \tag{1.14}$$

At any node (location where two or more paths/wires meet), the current in must be equal to the current out (of said node)

Example 1.9.1 Single Loop Example



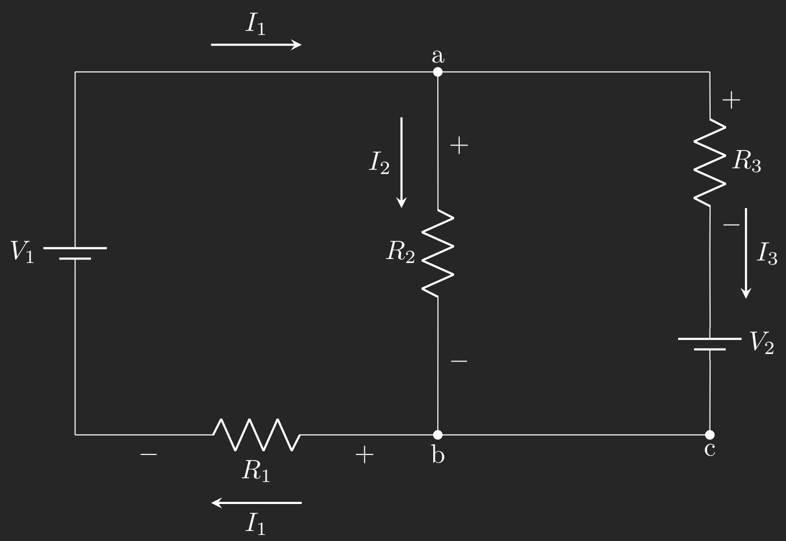
Can be simply solved by using Ohms law and doing $I = \frac{V_b}{R_1+R_2}$, or using Kirchhoff

Using Kirchhoff's Voltage Rule (1.13) we find the drop in current between $a \to b \to c \to V_b$

$$\Delta V_1 = +IR_1, \quad \Delta V_1 = +IR_2, \quad \Delta V_1 = -V_b$$

$$IR_1 + IR_2 - V_b = 0 \quad \Rightarrow \quad I = \frac{V_b}{R_1 + R_2}$$

Example 1.9.2 Two Loop Example



Using Kirchhoff's Current Rule(1.14) we find that $I_1 + I_2 + I_3 = 0$. To solve this we need equations with I_1, I_2, I_3

Using Kirchhoff's Voltage Rule (1.13) on the *outer* loop we get

$$I_3 R_3 + V_2 + I_3 R_1 - V_1$$

On the inner/left loop

$$I_2 R_2 + I_1 R_1 - V_1$$

Chapter 2

disc

2.1 pset1

2.1.1 a

It is possible that the third option c could be true. This is because we see an exponential acceleration in the Y direction, which correlates with W^2

2.1.2 b

$$a = F \quad t = \frac{L}{V_0} \rightarrow y_f = \frac{1}{2}\frac{FL^2}{V_0}$$

2.1.3 c

$$\frac{w}{2} = \frac{1}{2}\frac{F_{max}L^2}{mv_0^2} \qquad F_{max} = \frac{mv_0^2 w}{L^2}$$

2.1.4 d

$$tan(\theta) = \frac{v_y}{v_o} = \frac{w}{l}$$

$$v_y = \frac{F}{m} \times \frac{L}{v_0}$$

2.2

2.2.1

It has a net force of 0. This is because the x and y components cancel each other out.

2.2.2

The y component would cancel, as it is symmetric around the particle. X would not form a triangle to find r to use $u_g = \frac{-Gm_1m_2}{r^2}$

$$F_g = -\frac{GMm}{\left(\frac{\sqrt{5}a}{2}\right)^2}$$

$$F_{g,x} = -\frac{16Mm}{g\sqrt{g}a^2}$$

2.2.3

2.2.4

The physical argument is that the object is quite far away, and thus we can use a small angle aprox.

$$\sqrt{(x-a/2)^2+(a/2)^2} \rightarrow \approx x$$

$$\frac{GMm}{x^2}$$

2.3 RL stuff lol

it is the fourth one

2.3.1

$$u_c = -\frac{4\sqrt{2}GMm}{a}$$

2.3.2

$$\Delta u = \frac{4\sqrt{2}GMm}{a} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{8\sqrt{2}GM}{a}}$$

2.3.3

$$u_1, u_2 = -\frac{2GMm}{a}$$

$$u_3, u_4 = \frac{-2GMm}{\sqrt{5}a}$$

$$u = u_1 + u_2 \cdots = -\frac{4GMm}{a}(1 + \frac{1}{\sqrt{5} = Ur})$$

2.3.4

no, the middle has *higher* potential energy

2.3.5

2.4

2.4.1

$$p^2 = \frac{4a^3\pi^2}{GM}$$

2.4.2

$$\omega = 2\pi/p \rightarrow \omega^2 = \frac{\cancel{4\pi^2}GM}{\cancel{4}a^3\cancel{\pi^2}} = \frac{GM}{r^3}$$

2.4.3

$$Dw^a R6a + b$$

$$a = \frac{D}{m}w^a R^{a+b} = \omega^2 R$$

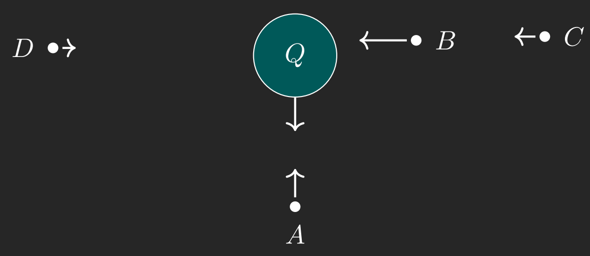
$$\omega^{2-a} = \frac{D}{m}R^{a+b-1}$$

2.4.4

2.5 w2

2.5.1 q1

Example 2.5.1 The configuration below shows some points around a positive charge Q. If a positive test charge q is placed at point A, at a distance rA from Q, calculate the magnitude of the force between the charges in terms the given variables.



P1) Find magnitude

Because we're finding a charge diff, we use 1.1

$$\vec{F_{12}} = k\frac{Qq}{r_A^2}$$

P2) Force strength

Based on distance to the *field* of Q

$$F_B > F_A > F_C > F_D$$

P4) Electric Field Mag

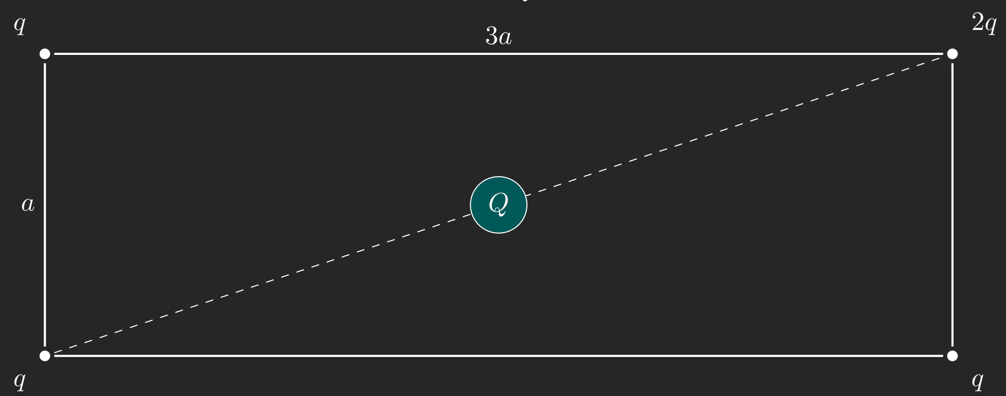
Because a test charge placed in a field exerts doesn't have any force of its own, we use a modified version of 1.3

$$E = k\frac{Q}{r_a^2}$$

Based on above...

$$2Q \propto 2E, \quad 2q \not\propto 2E$$

Example 2.5.2 Electric Field Due to Point Charges Four point charges 2q, q, q, and q are placed at the corners of a rectangle of dimensions a and 3a as shown in the figure. A fifth charge Q is placed at the center of the rectangle. Our task is to compute the electric field at the center of the rectangle, and then determine the force on Q.



P1) Find the mag of field charges

$$r = \sqrt{\left(\frac{3a}{2}\right)^2 + \frac{1}{(2a)^2}}$$

$$r = \sqrt{\frac{9a^2}{4} + \frac{1}{4a^2}}$$

$$r = \sqrt{\left(\frac{10a^2}{4}\right)}$$

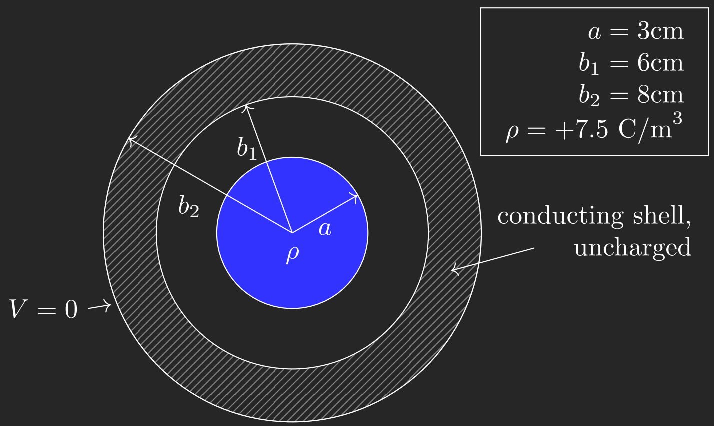
$$r = \frac{\sqrt{5}}{\sqrt{2}}a$$

$$q = E_1 = \frac{2ka}{5a^2}, \quad 2q = E_2 = \frac{4ka}{5a^2}$$

P3) Find X and Y components

2.6 D4

Example 2.6.1 Electric Potential in a System with Cylindrical Symmetry



Find the potential at r = 10cm

Find the energy enclosed using Gauss's Law (1.6)

$$\int \vec{E}dS = \frac{Q_{enc}}{\epsilon_0}$$

$$\lambda = \rho * \pi * a^2$$

$$Q_{enc} = \rho \pi a^2 L$$

$$E(2\cancel{\pi}r) = \rho\cancel{\pi}a^2$$

$$E = \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \int_r^{b_2} \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b_2}{r}\right)$$

2.7 Equations

$$\vec{F_{12}} = k\frac{q_1 q_2}{r_{12}^2}\hat{r_{12}} \quad (1.1)\ (\textit{Coulomb's Law})$$

$$F_{net} = \sum_i \vec{F_i} \quad (1.2)\ (\textit{Superpostion Principle})$$

$$E = k\frac{q}{r^2} \quad (1.3)\ (\textit{Electric Field})$$

$$D = \frac{N}{4\pi r^2} \quad (1.4)\ (\textit{Density of Field Lines})$$

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \quad (1.5)\ (\textit{Number of Graphed Field Lines })$$

$$\int EdS = \frac{Q_{enclosed}}{\epsilon_0} \quad (1.6)\ (\textit{Guass's Law})$$

$$\Delta U = q\vec{E}\ell \quad (1.7)\ (\textit{Change in potential in const E field})$$

$$\Delta U = Q\int_a^b \vec{E}dr \quad (1.8)\ (\textit{Change in potential in non static fie}$$

$$C \equiv \frac{Q}{\Delta V} \quad (1.9)\ (\textit{Capacitance})$$

$$C = \frac{\epsilon_0 A}{d} \quad (1.10)\ (\textit{Capacitance for Parrel Plates})$$

$$U = \frac{1}{2}\frac{Q^2}{C} \quad U = \frac{1}{2}CV^2 \quad U = \frac{1}{2}QV \quad (1.11)\ (\textit{Potential of a Capacitor})$$

$$u = \frac{1}{2}\epsilon_0 E^2 \quad (1.12)\ (\textit{Energy density of a capacitor})$$

$$\sum \Delta V_n = 0 \quad (1.13)\ (\textit{Kirchhoff Voltage Rule})$$

$$\boxed{\sum I_{in} = \sum I_{out}}, \text{at any node} \quad (1.14)\ (\textit{Kirchhoff Current Rule})$$

2.8 Constants

$$e = -1.6 \times 10^{-19} C \quad (1)\ (\textit{Electron Charge})$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2} \quad (2)\ (k)$$

Glossary

Dielectric Constant Denoted as κ
$C_{new} = \kappa C_{old}$
The amount which the capacitance of a capacitor increases when a dielectric is placed between the charges sufraces 13